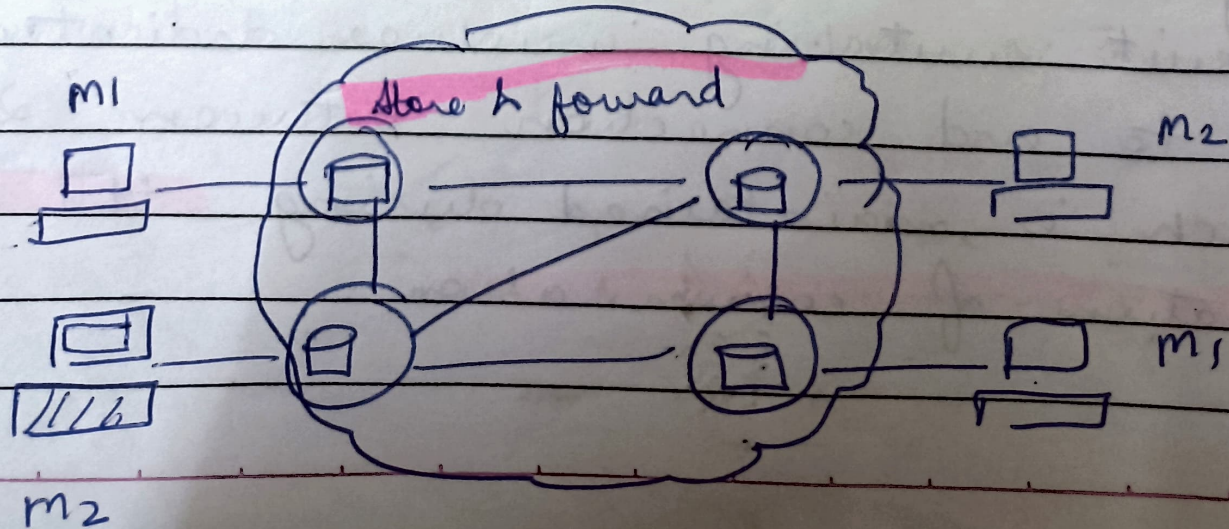


MESSAGE

SWITCHING



In message switching, each message is treated as an independent unit and includes its own destination and source address.

source $\rightarrow$ $\rightarrow$ destination

SWITCHES ARE INTERCONNECTED BY TRUNKS

Every intermediate device receives the message stores it until the next device is ready to receive it and forwards it to the next device.

It is also called as a store and forward network

ADVANTAGES:

- 1) Efficient traffic management by assigning priorities to the messages to be switched.
- 2) Reduced traffic because it can store a message until communication channel is available
- 3) Data channels are shared.

DISADVANTAGES:

- 1) The storing and forwarding introduces delay and hence cannot be used for real time applications like voice & video
- 2) Storage is needed for intermediate devices

DELAY: Switch needs buffer

FEATURES

- 1) Message switching is unidirectional. After a message is sent no acknowledgement is sent.
- 2) Error control mechanisms like error checking bits, etc.
- 3) Switch to switch transfer is an independent operation. Links (connection between switches) operate at different data rates.
- 4) Every message is an independent entity so address is written on every message.

STRUCTURE OF A SWITCH

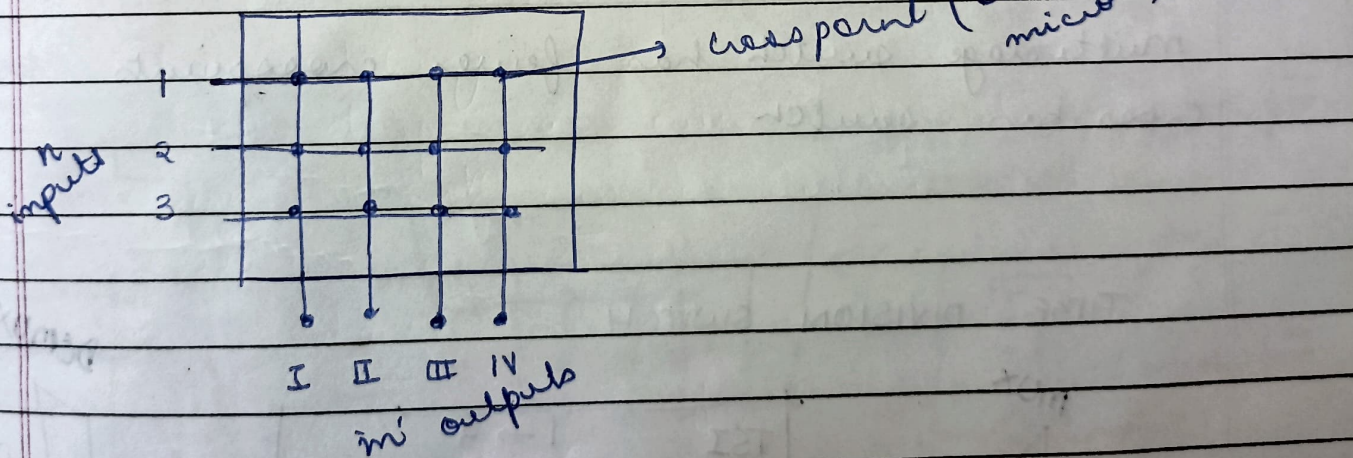
Switching uses 2 technologies

- 1) Space division switch
- 2) Time division switch

SPACE DIVISION SWITCH

The paths in the circuit are separated from one another spatially. It is being used in both analog & digital networks.

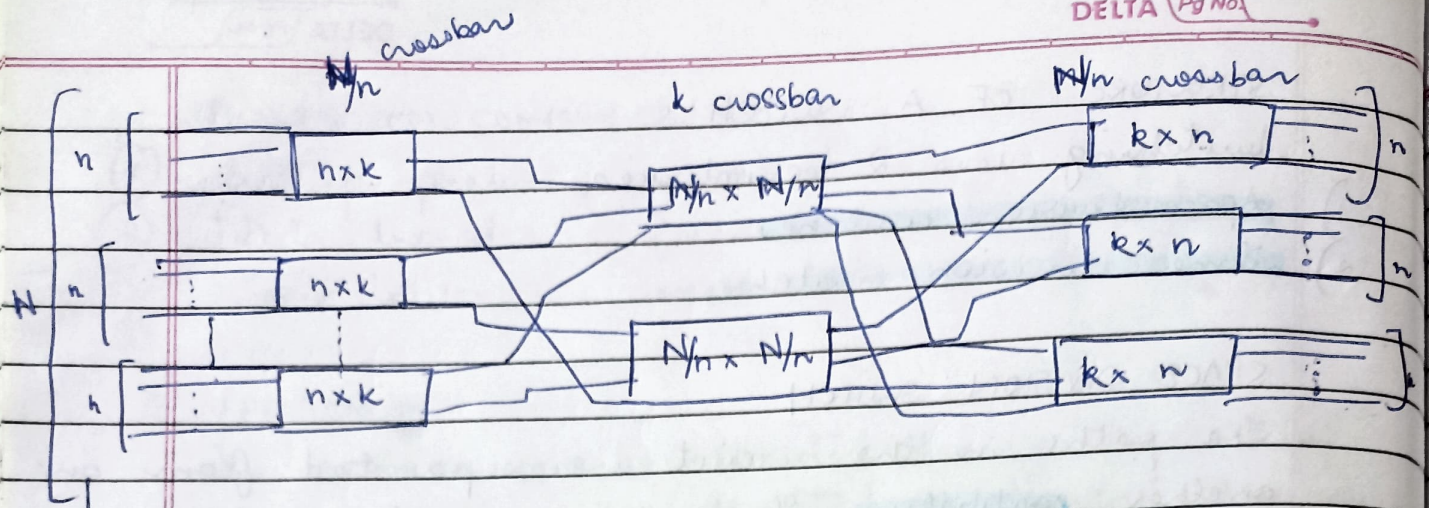
- ① CROSSBAR SWITCH: It connects n inputs to m outputs in a grid using electronic microswitches at each crosspoint. The major drawback is no. of crosspoints required. ' n ' inputs ' m ' outputs $n \times m$ crosspoints. Impractical



MULTISTAGE SWITCH: It is a combination of crossbar switches in several stages. By allowing multiple paths inside a switch we can reduce the no. of cross points.

MULTI STAGE SWITCH

Date 28/8/23
DELTA Pg No.

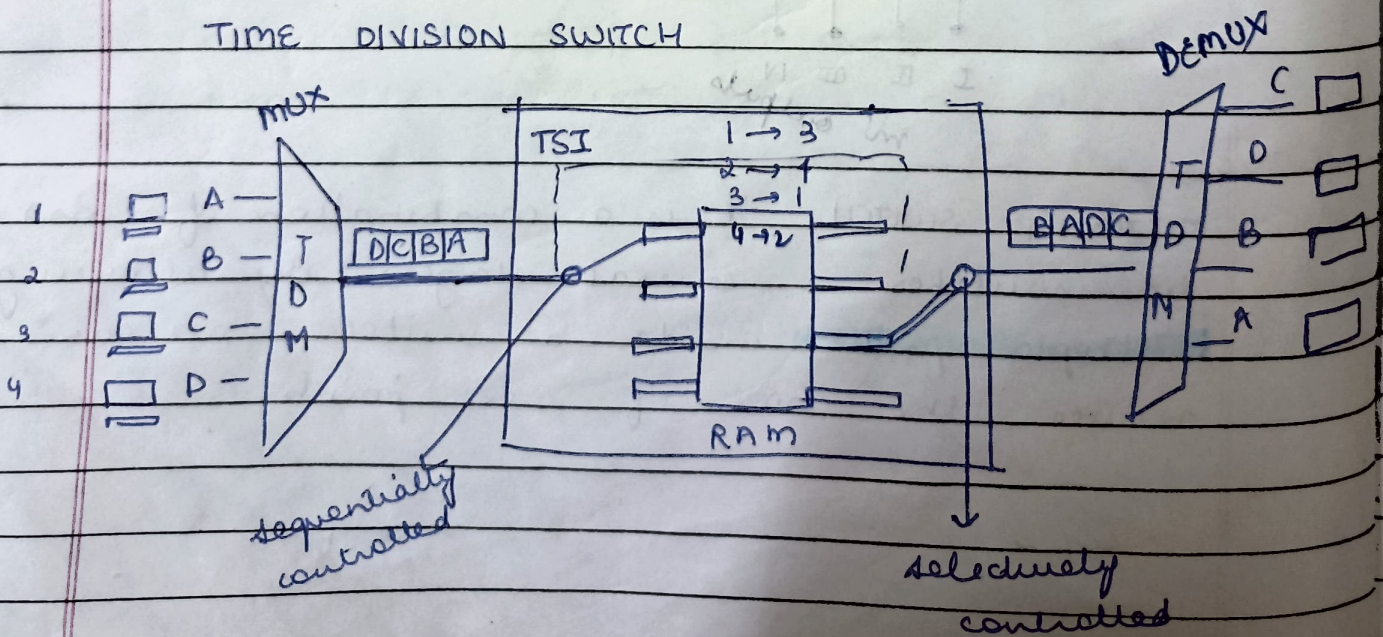


- ① Input (N) lines are divided into groups each of 'n' lines
- ② For every group we use crossbar of size $n \times k$
 $k \rightarrow$ no. of crossbars in middle stage

The first stage has N/n cross bars and $n \times k$ crosspoints.

Multi stage switch has fewer crosspoints than cross bar switch

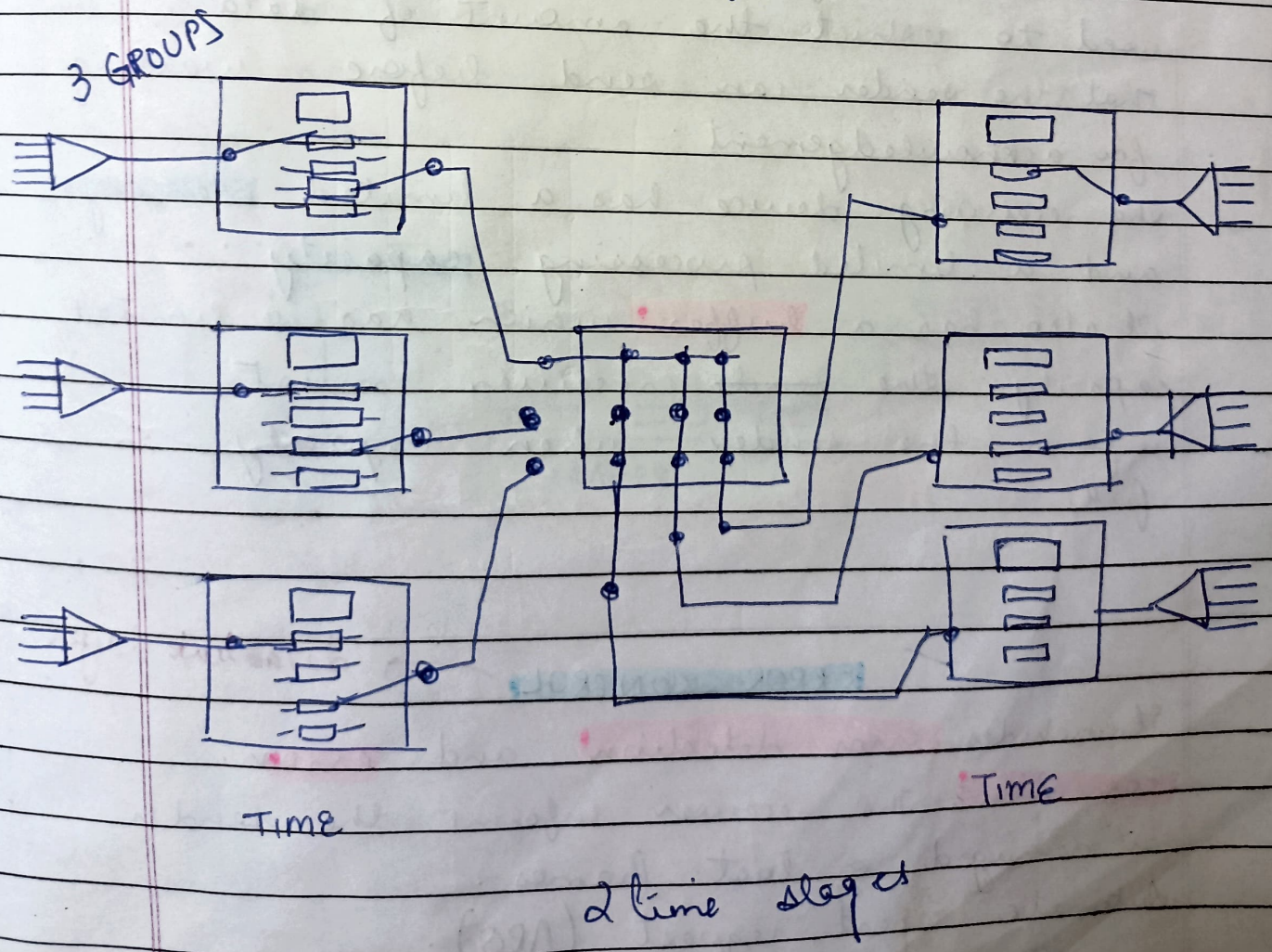
TIME DIVISION SWITCH



- ① We have a TDM multiplexer, a TDM demultiplexer and a TSI consisting of RAM with several memory locations.
- ② RAM fills up with incoming data from line slots. Slots are sent out in order based on control unit.

TIME & SPACE DIVISION SWITCH COMBINATION
 Space division is instantaneous but uses crosspoints. Time division has no crosspoints but RAM causes delay.

TSI switch uses best of both



We have a TIME SPACE TIME TST switch which consists of 2 time stages and one space stage. We have 12 inputs & 12 outputs. The inputs are divided into 3 groups.

The average delay is minimized.
Cross points are minimized.

Data Link Layer

Unit II

FLOW CONTROL

It coordinates the amount of data that can be sent before receiving an acknowledgement and is an important duty of data link layer.

Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgement.

The receiving device has a limited memory and a limited processing capacity.

It also has a buffer which has a limited capacity. The sender/receiver must inform the sender when capacity is full.

ERROR CONTROL

→ Data Link Layer

It includes error detection and error correction. The receiver informs the sender of damaged or lost frames: Automatic repeat request (ARQ).

Retransmission of frames

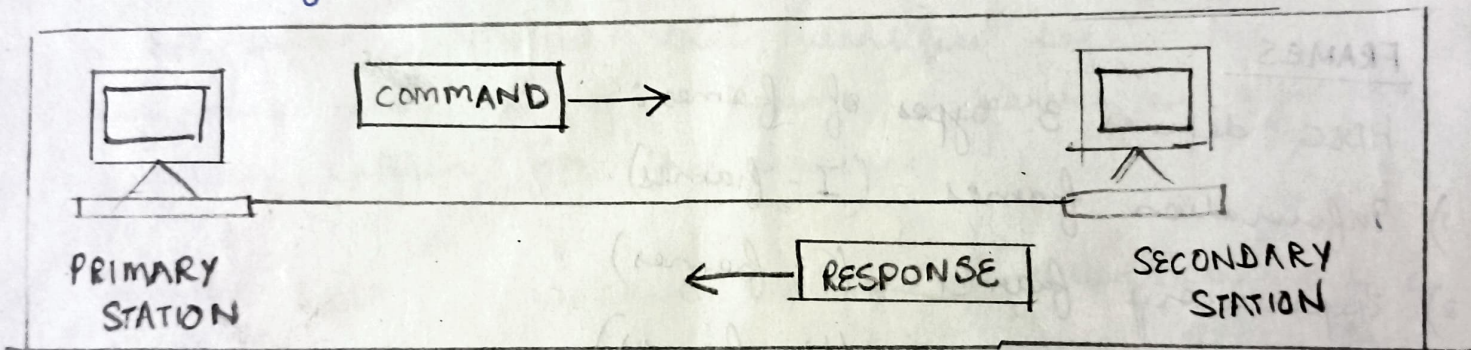
HDLC

HDLC stands for High level data link control. It is a bit oriented protocol for communicating over point to point and multi-point links.

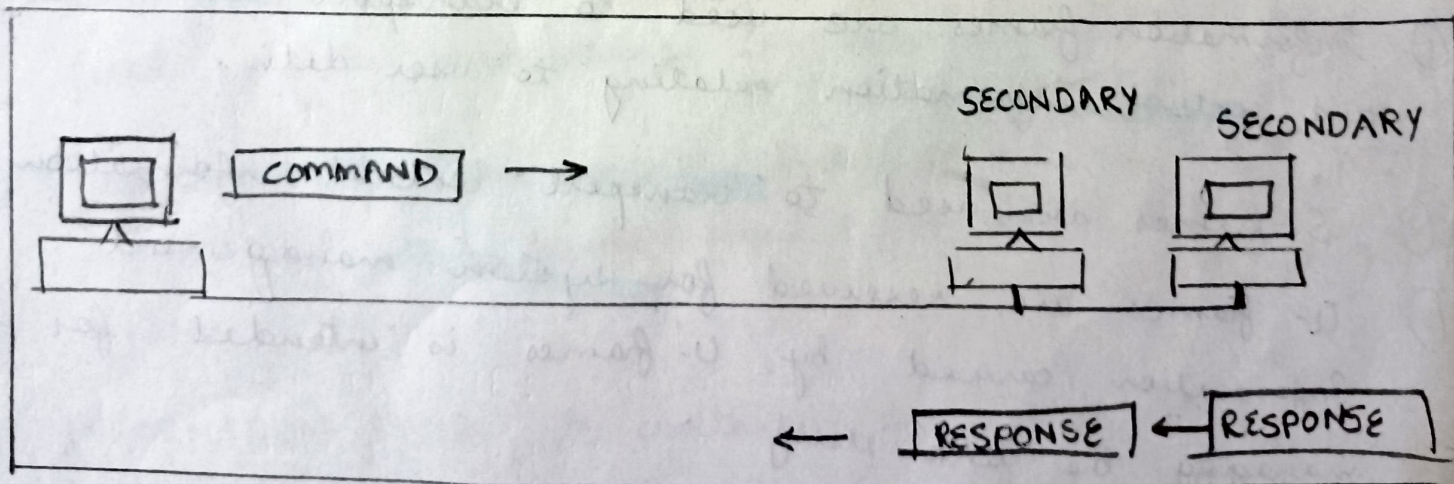
It provides 2 common transfer modes:

- 1) Normal Response Mode (NRM)
- 2) Asynchronous Balanced mode (ABM)

① NORMAL RESPONSE MODE: In NRM, the station configuration is unbalanced. We have one primary station and multiple secondary stations. A primary station can send commands and a secondary station can only respond.



(a) POINT TO POINT



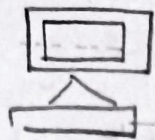
(b) MULTIPOINT

NORMAL RESPONSE MODE

② ASYNCHRONOUS BALANCED MODE :

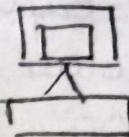
In ABM configuration is balanced. The link is point to point and each station can function as primary and secondary station.

COMBINED



COMMAND/RESPONSE

COMBINED



COMMAND/RESPONSE

ASYNCHRONOUS BALANCED MODE

FRAMES

HDLc defines 3 types of frames:

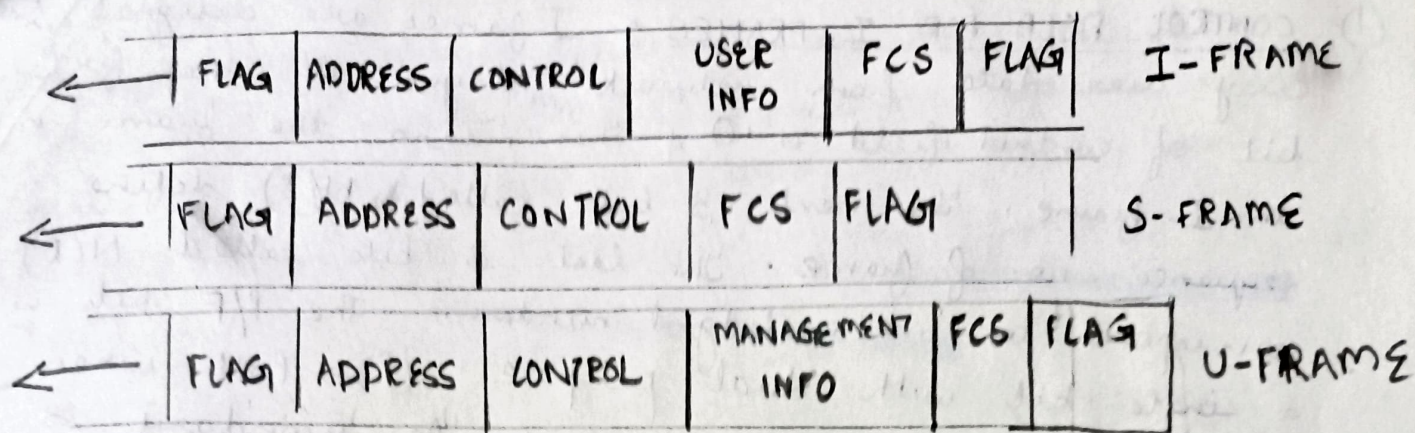
- 1) Information frames (I-frames)
- 2) Supervisory frames (S-frames)
- 3) Unnumbered frames (U-frames)

1) Information frames are used to transport user data and control information relating to user data.

2) S-frames are used to transport control information.

3) U-frames are reserved for system management. Information carried by U-frames is intended for managing the link itself.

FRAME FORMAT

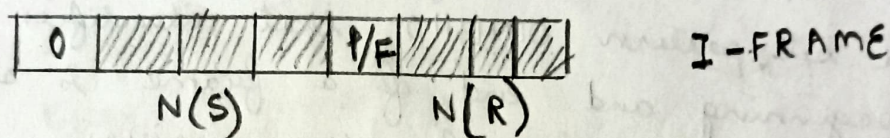


FIELDS

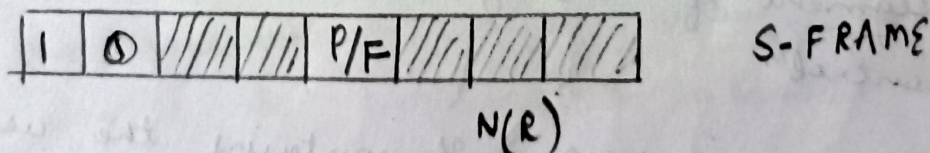
- ① FLAG FIELD:** The flag field is an 8-bit sequence with a bit pattern 01111110 that identifies both the beginning and end of a frame & serves as synchronization pattern for the receiver.
- ② ADDRESS FIELD:** It contains address of secondary station. If a primary station created the frame, it contains a to address and if a secondary station created the frame it contains from address. An address field can be one byte or several bytes long.
- ③ CONTROL FIELD:** The control field is a one or 2-bytes segment of the frame used for flow & error control.
- ④ INFORMATION FIELD:** It contains the users data from the network layer. The length of this field varies from network to network.
- ⑤ FCS FIELD:** The frame check sequence FCS is the HDLC error detection field.

CONTROL FIELD

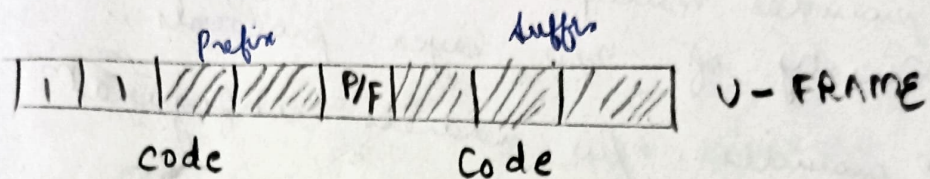
- ① CONTROL FIELD FOR I-FRAMES: I frames are designed to carry user data from network layer. If the first bit of control field is 0, this means the frame is an I-frame. The next 3 bits called $N(S)$ define sequence no. of frame. The last 3 bits called $N(R)$ correspond to acknowledgment number. The P/F bit is a single bit with dual purpose. (P) $\rightarrow$ Poll when the frame is sent by primary to secondary & 'F' final when frame is sent from secondary to primary -



- ② CONTROL FIELD FOR S FRAMES: Supervisory frames are used for error and flow control whenever piggybacking is impossible. No information field is present. If first 2 bits of control field is 10, it means the frame is an S-frame. The last 3 bits $N(R)$ correspond to acknowledgment number or NAK (negative acknowledgment no.).



CONTROL FIELD FOR U-FRAMES: Unnumbered frames are used to exchange session management and control information between connected devices. The info management field manages system information. We have a 2-bit prefix before P/F field, and 3-bit suffix after P/F field. The frame starts with 11.



POINT TO POINT PROTOCOL

It is a very common protocol for point to point access. Nowadays millions of computers use PPP protocol to connect their home computers to the server of an ISP. They are connected to the internet through a telephone line which provides the services of physical layer.

It provides services like,

- 1) PPP defines the format of the frame to be exchanged between the devices.
- 2) It defines how 2 devices can negotiate the establishment of link & exchange of data.
- 3) It defines how the network layer data is encapsulated in the data link frame.
- 4) It defines how 2 devices can authenticate with each other.
- 5) It provides connections over multiple links.

one PPP is bit

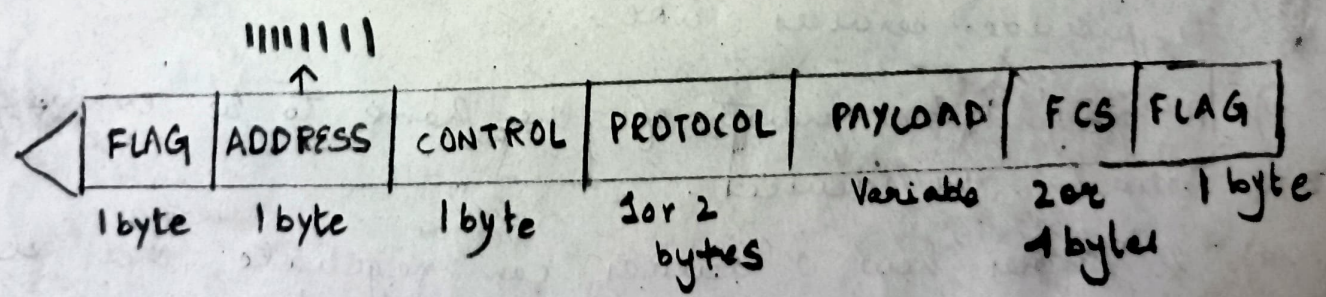
~~the following are the main features of PPP~~

- ⑥ It provides multiple network layer services supporting a variety of n/w layer protocols.
- ⑦ PPP provides n/w address configuration.

Some of the services missing in PPP are:

- 1) PPP does not provide flow control
- 2) Error control mechanism is simple. It uses CRC (cyclic redundancy check)
- 3) It does not provide a sophisticated addressing mechanism

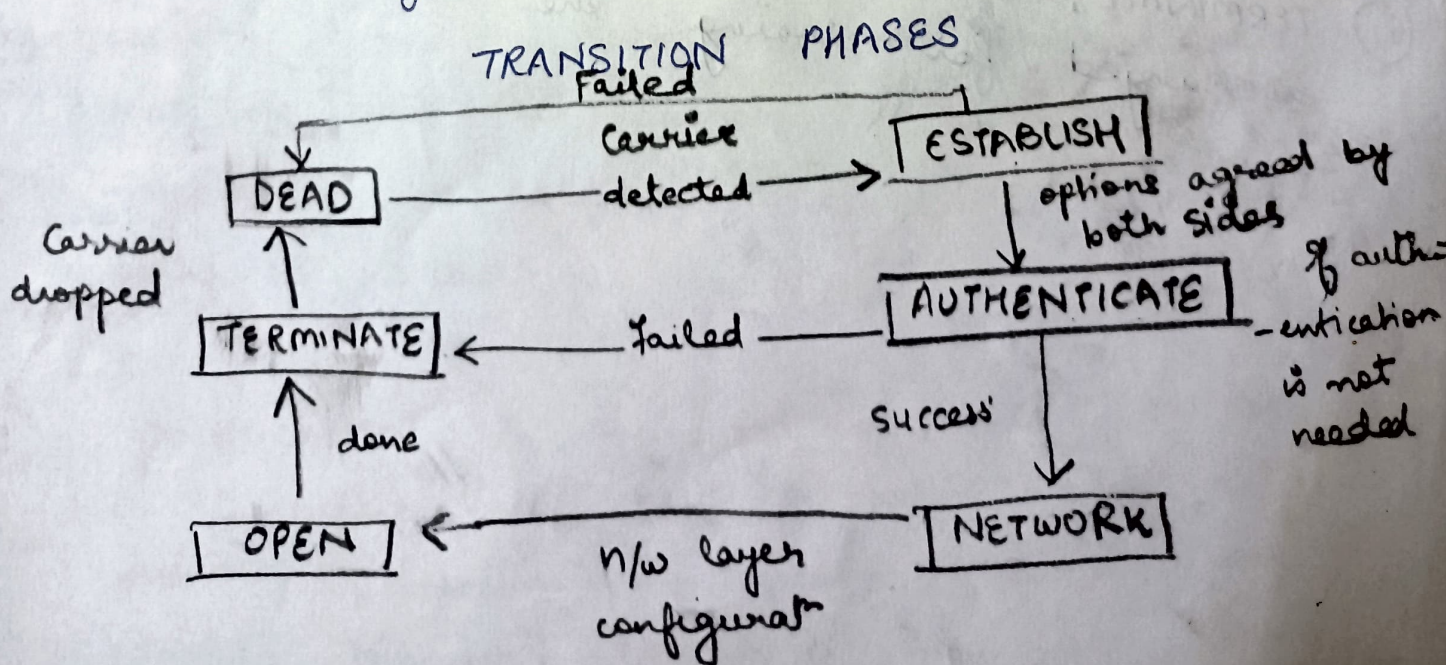
FRAMING : PPP is a byte oriented protocol



PPP FRAME FORMAT

LAG: A PPP frame starts and ends with a one byte flag with a bit pattern 01111110. PPP is a byte oriented protocol, HDLC is a bit oriented protocol.

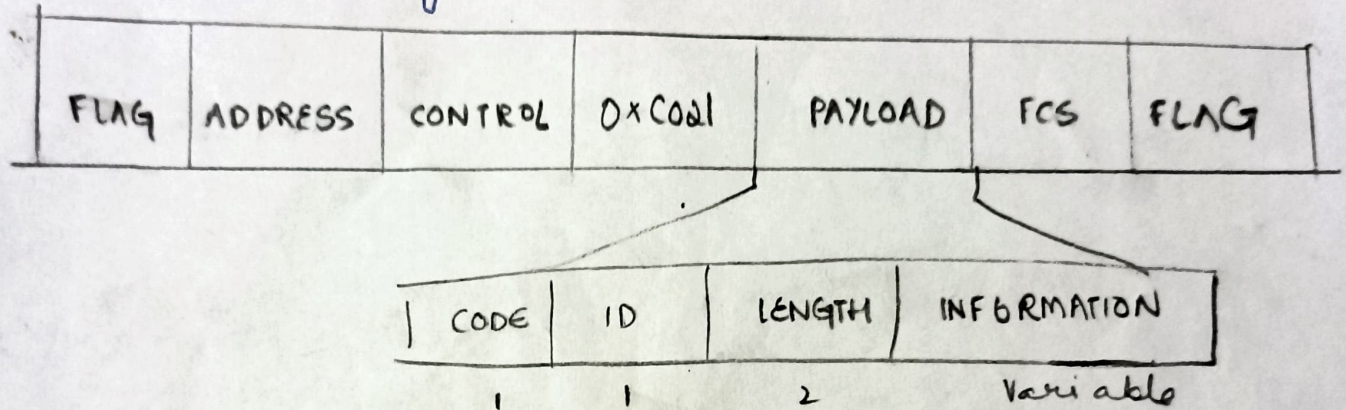
- ② ADDRESS: The address field is a constant value and is set to 1111111.
- ③ CONTROL: This field is set to a constant value of 11000000. PPP doesn't provide flow control.
- ④ PROTOCOL: It defines the type of data being carried in data field, user data or other form of data. It can be 1 or 2 bytes long.
- ⑤ PAYLOAD FIELD: The field either carries user data or other information.
- ⑥ FCS: The frame check sequence (FCS) is a 2 or 4 byte standard CRC.



- 1) DEAD: In the dead phase the link is not used. The line is quiet.
- 2) ESTABLISH: When one of the nodes starts the communication, the connection goes into this phase. In this phase, options are negotiated between 2 parties. If negotiation is successful, the system goes into authentication phase or directly to network phase.
- 3) AUTHENTICATE: This phase is optional, it may be skipped, but if the parties proceed, they have to send several authentication packets.
- 4) NETWORK: In this phase negotiation for network layer protocols take place. An agreement is established before data at network layer can be exchanged.
- 5) OPEN: Data transfer takes place, exchange of packets is done.
- 6) TERMINATE: Connection is terminated. Packets are exchanged for closing the link.

- the ^{base} _{gen}
- STACK: Three set of protocols make PPP powerful
- 1) Link Control Protocol
 - 2) Authentication protocol
 - 3) Network Control protocols

① LINK CONTROL PROTOCOL: It is responsible for establishing, maintaining, configuring and terminating links. It also provides negotiation options.



- (i) **CODE**: It is a one byte length field which defines type of LCP packet.
- (ii) **ID**: It is a one byte length field which holds a value used to match a request with reply.
- (iii) **LENGTH**: It is a 2 byte long field which defines length of entire LCP packet.
- (iv) **INFORMATION**: It holds extra information needed for some LCP packets.

② **AUTHENTICATION**: It plays a very important role in PPP where verification of user identity is necessary. Authentication means validating the identity of user who needs to access some resources. It uses (PAP) Password Authentication Protocol.

③ NETWORK CONTROL PROTOCOLS: It defines a specific protocol for each network Control protocol for each set of protocols.